

PROF MAIBVISE

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INTEGRATION BY PARTS EXPLAINED

We use the concept of integration by parts when we want to integrate product of different functions for example

$$\int e^{2x} \cos 2x \, dx \quad \int x \ln x \, dx \quad \int 2x e^{2x} \, dx \text{ etc}$$

We see that there is product of an exponential and trig function in the first example, product of algebraic function and log in the second and product of an algebraic and an exponential function in the last.

So when we want to integrate the product of different functions we now use the Integration by Parts Formula.

The formula for Integration by parts used to integrate the products of 2 functions of x is as follows:

Let us derive it from the product rule:

$$\text{if } y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Now make $u \frac{dv}{dx}$ subject of formula

$$u \frac{dv}{dx} = \frac{dy}{dx} - v \frac{du}{dx}$$

Integrating both sides gives w.r.t x

$$\int \left(u \frac{dv}{dx} \right) dx = \int \left(\frac{dy}{dx} - v \frac{du}{dx} \right) dx$$

$$\int \left(u \frac{dv}{dx} \right) dx = \int \left(\frac{dy}{dx} \right) dx - \int \left(v \frac{du}{dx} \right) dx$$

$$\int \left(u \frac{dv}{dx} \right) dx = y - \int \left(v \frac{du}{dx} \right) dx$$

Remember:

$$\int \left(\frac{dy}{dx} \right) dx = y$$

$$\int \left(u \frac{dv}{dx} \right) dx = uv - \int \left(v \frac{du}{dx} \right) dx$$

remember $y = uv$

∴ our formula for integration by parts goes as follows

$$\int \left(u \frac{dv}{dx} \right) dx = uv - \int \left(v \frac{du}{dx} \right) dx$$

Product of the two functions

Between those two functions multiplying each other there is one which should be integrated and the other differentiated

THE PROBLEM IS HOW DO WE KNOW THE ONE TO INTEGRATE AND THE ONE TO DIFFERENTIATE

We determine the one to be differentiated through the acronym

LATE

The order of differentiation is as follows

Logarithmic Function

Algebraic Function

Trigonometrical function

Exponential Function

Once we see the one to differentiate it will be obvious that the remaining should be integrated

Let us do the following examples together

$$(a) \int x \cos x \, dx$$

We need to know which one to integrate or differentiate by using

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We see that Algebraic expressions (in this example x) must be differentiated first before Trigs (in this example $\cos x$)

$\therefore$ we are going to differentiate x and integrate $\cos x$

The one to differentiate is equated to u and gives $\frac{du}{dx}$

The one to integrate is equated to $\frac{dv}{dx}$ and gives v

$$\therefore u = x \qquad \frac{dv}{dx} = \cos x$$

$$\frac{du}{dx} = 1$$

$$v = \int \left(\frac{dv}{dx} \right) dx$$

$$v = \int \cos x \, dx$$

$$v = \sin x$$

$$\int x \cos x \, dx = (x)(\sin x) - \int (\sin x)(1) \, dx$$

$$\int \left(u \frac{dv}{dx} \right) dx = uv - \int \left(v \frac{du}{dx} \right) dx$$

$$\begin{aligned} \int x \cos x \, dx &= x \sin x - \int (\sin x) \, dx \\ &= x \sin x - (-\cos x) \end{aligned}$$

$$= x \sin x + \cos x + c$$

Differentiate (u)

(2) $\int x \ln x \, dx$ By using LATE

Integrate $\left(\frac{dv}{dx} \right)$

Log differentiated first!!!

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\frac{dv}{dx} = x$$

$$v = \int x dx$$

$$v = \frac{x^2}{2} + c$$

$$\therefore \int x \ln x \, dx = (\ln x) \left(\frac{1}{2} x^2 \right) - \int \left(\frac{1}{2} x^2 \right) \left(\frac{1}{x} \right) dx$$

$$= \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x \, dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{2} \int x \, dx$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{2} \left(\frac{1}{2} x^2 \right) + c$$

$$= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + c$$

$$= \frac{1}{4} x^2 (2 \ln x - 1) + c$$

Special Cases for Integration by Parts

(1) The Integration of $\ln x$

Let us see why it is considered special

Consider $\int \ln x \, dx$

Do you see that there is product of two functions but the other one seems to be hidden~ maybe it has no confidence to show itself but actually it is 1!!!

$$\int \ln x \, dx = \int 1 \cdot \ln x \, dx$$

Now we know according to LATE that in all cases where there is a log, differentiate it and the other item integrate.

$$\therefore u = \ln x \quad \frac{dv}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{x} \quad v = x$$

$$\therefore \int \ln x \, dx = x \ln x - \int (x) \left(\frac{1}{x}\right) dx$$

$$= x \ln x - \int 1 \, dx$$

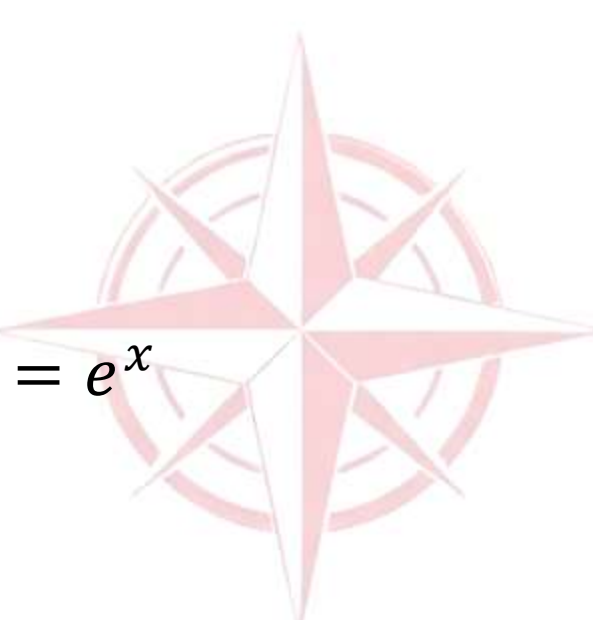
$$= x \ln x - x + c$$

(2) Integration of a trig and an exponential Function

Let us look at the following example and identify why it is considered special case.

$$\text{Find } \int e^x \sin x \, dx$$

By LATE

$$u = \sin x \quad \frac{dv}{dx} = e^x$$


$$\frac{du}{dx} = \cos x \quad v = e^x$$

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$$\therefore \int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx$$

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And $\int e^x \cos x \, dx$ needs by parts again

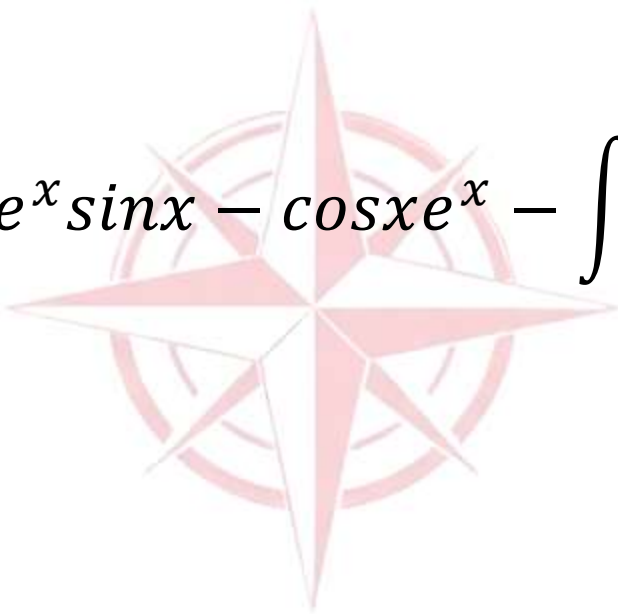
$$\int e^x \cos x \, dx = \cos x e^x - \int -\sin x e^x \, dx$$

$$\int e^x \cos x \, dx = \cos x e^x + \int \sin x e^x \, dx$$

Now substitute $\int e^x \cos x \, dx$ back

$$\therefore \int e^x \sin x \, dx = e^x \sin x - \left(\cos x e^x + \int \sin x e^x \, dx \right)$$

$$\therefore \int e^x \sin x \, dx = e^x \sin x - \cos x e^x - \int \sin x e^x \, dx$$



Now make $\int e^x \sin x \, dx$ Subject

$$\int e^x \sin x \, dx + \int e^x \sin x \, dx = e^x \sin x - \cos x e^x + c$$

$$2 \int e^x \sin x \, dx = e^x \sin x - \cos x e^x + c$$

$$\therefore \int e^x \sin x \, dx = \frac{1}{2} (e^x \sin x - \cos x e^x) + c$$

(3) Integrating By Parts Twice

Let us do the following example

$$\text{Find } \int x^2 e^{3x} dx$$

Using LATE

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$\frac{dv}{dx} = e^{3x}$$

$$v = \frac{1}{3} e^{3x}$$

$$\therefore \int x^2 e^{3x} = (x^2) \left(\frac{1}{3} e^{3x} \right) - \int \left(\frac{1}{3} \right) e^{3x} (2x) dx$$

$$= \frac{1}{3} x^2 e^{3x} - \int \frac{2}{3} x e^{3x} dx$$

$$\int x^2 e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int x e^{3x} dx$$

This needs by parts again!!!!

So taking that aside:

$$\int x e^x dx$$

Using LATE

$$u = x \qquad \frac{dv}{dx} = e^x$$

$$\frac{du}{dx} = 1 \qquad v = e^x$$

$$\begin{aligned} \therefore \int x e^x dx &= (x)(e^x) - \int e^x (1) dx \\ &= x e^x - e^x + c \end{aligned}$$

So we now then substitute back in

$$\int x^2 e^{3x} dx = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int x e^{3x} dx$$

Where there is $\int x e^{3x} dx$ we are putting $x e^x - e^x$ to get

$$\int x^2 e^{3x} = \frac{1}{3} x^2 e^{3x} - \frac{2}{3} (x e^x - e^x) + c$$

$$= \frac{1}{3} x^2 e^{3x} - \frac{2}{3} x e^x + \frac{2}{3} e^x + c$$

$$= \frac{1}{3} e^{3x} (x^2 - 2x + 2) + c$$

To Do Questions

Integrate the following w.r.t x

$$(1) \int x e^x dx \quad (2) \int x \cos x dx \quad (3) \int x^2 \ln x dx$$

$$(4) \int x^2 \cos x dx \quad (5) \int x e^{4x} dx$$

Short Hints for Solutions

$$(1) u = x \quad \text{and} \quad \frac{dv}{dx} = e^x \quad [\text{Ans: } x e^x - e^x + c]$$

$$(2) u = x \quad \frac{dv}{dx} = \cos x \quad [\text{Ans: } x \sin x + \cos x + c]$$

$$(3) u = \ln x \quad \frac{dv}{dx} = x^2 \quad \text{Ans: } \left[\frac{x^3}{3} \ln x - \frac{x^2}{4} + c \right]$$

$$(4) u = x^2 \quad \frac{dv}{dx} = \cos x \quad [\text{Ans: } x^2 \sin x + 2x \cos x - 2 \sin x]$$

$$(5) u = x \quad \frac{dv}{dx} = e^{4x} \quad \left[\text{Ans: } \frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} + c \right]$$

Stay Blessed

Professor



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